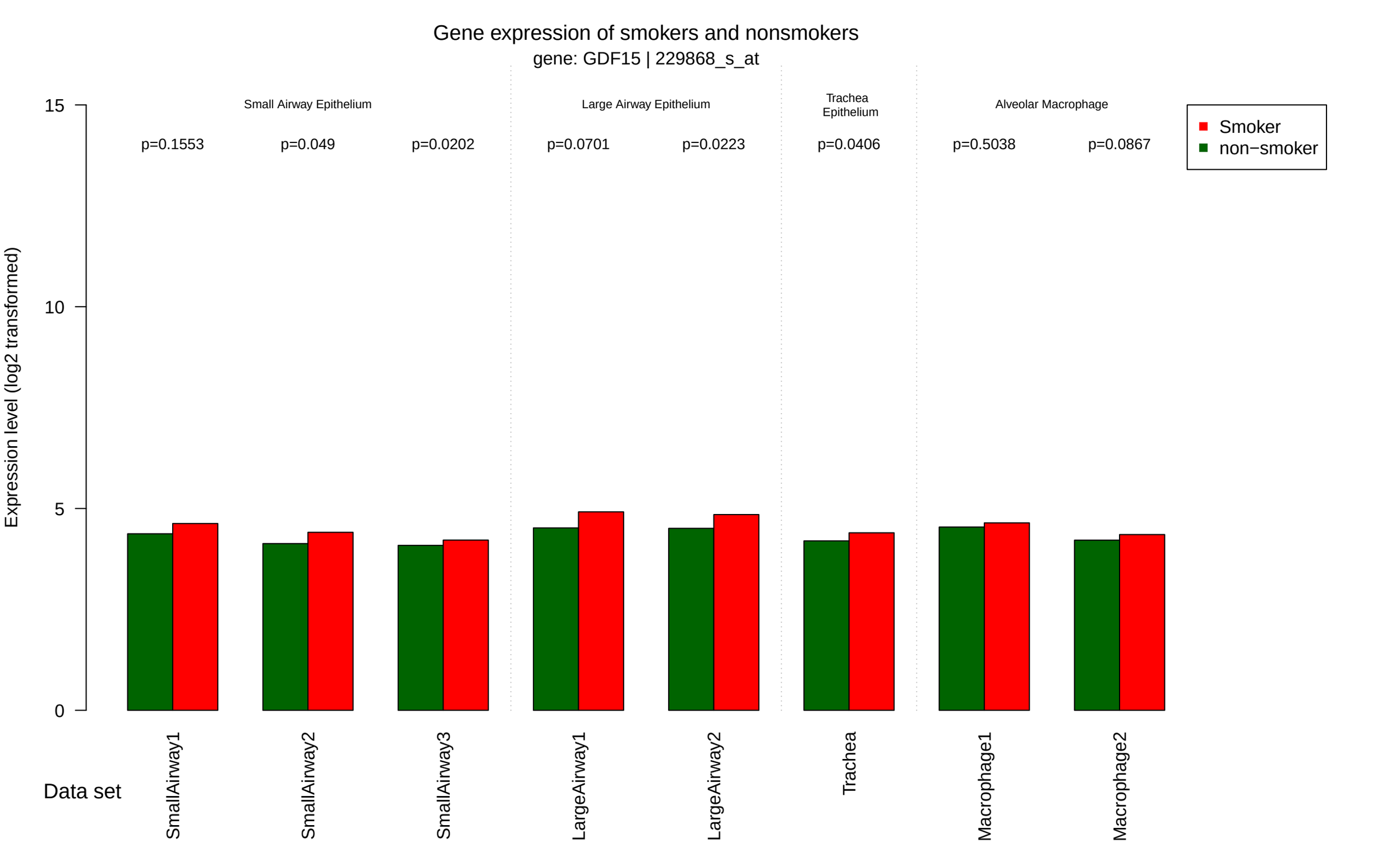
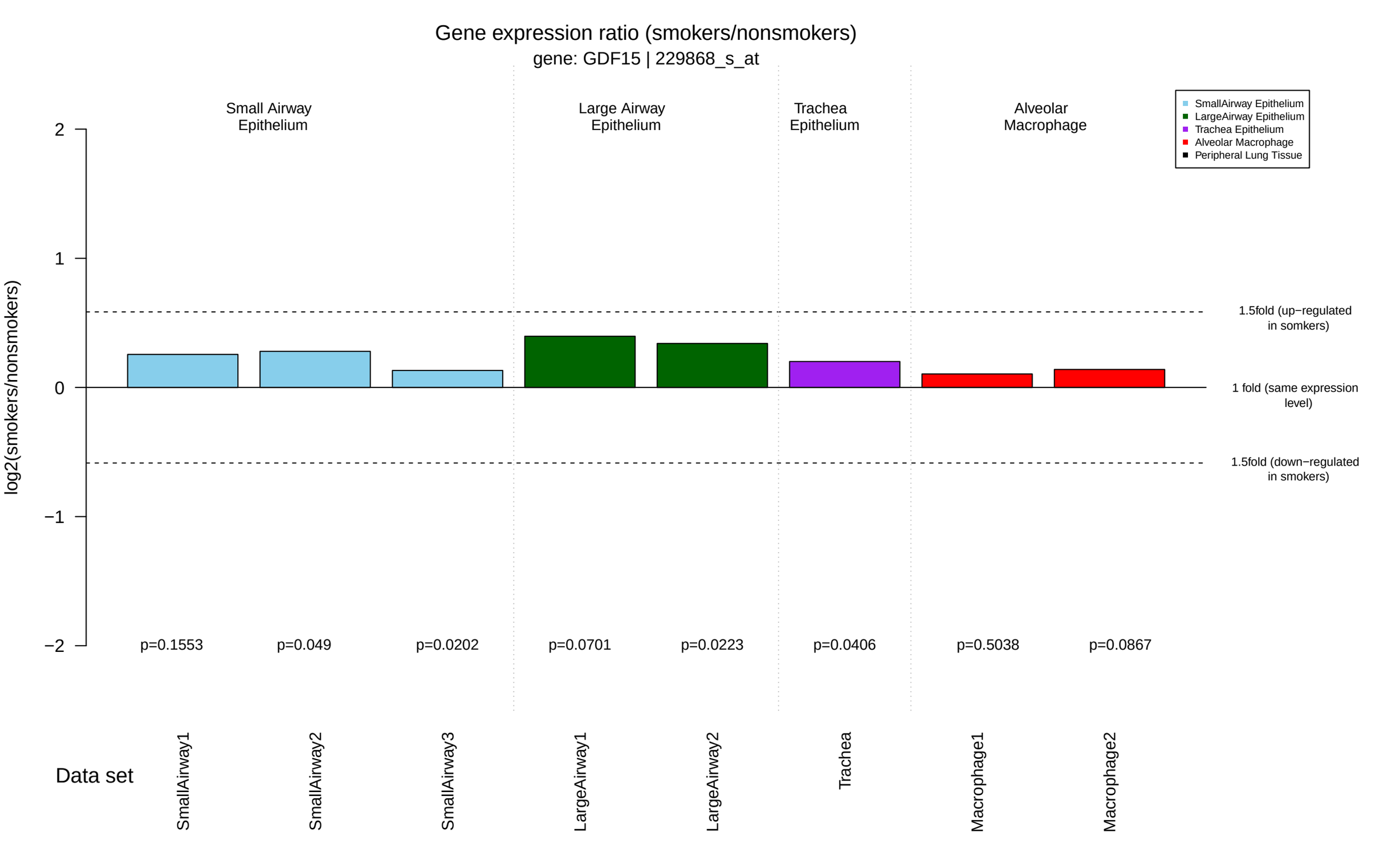
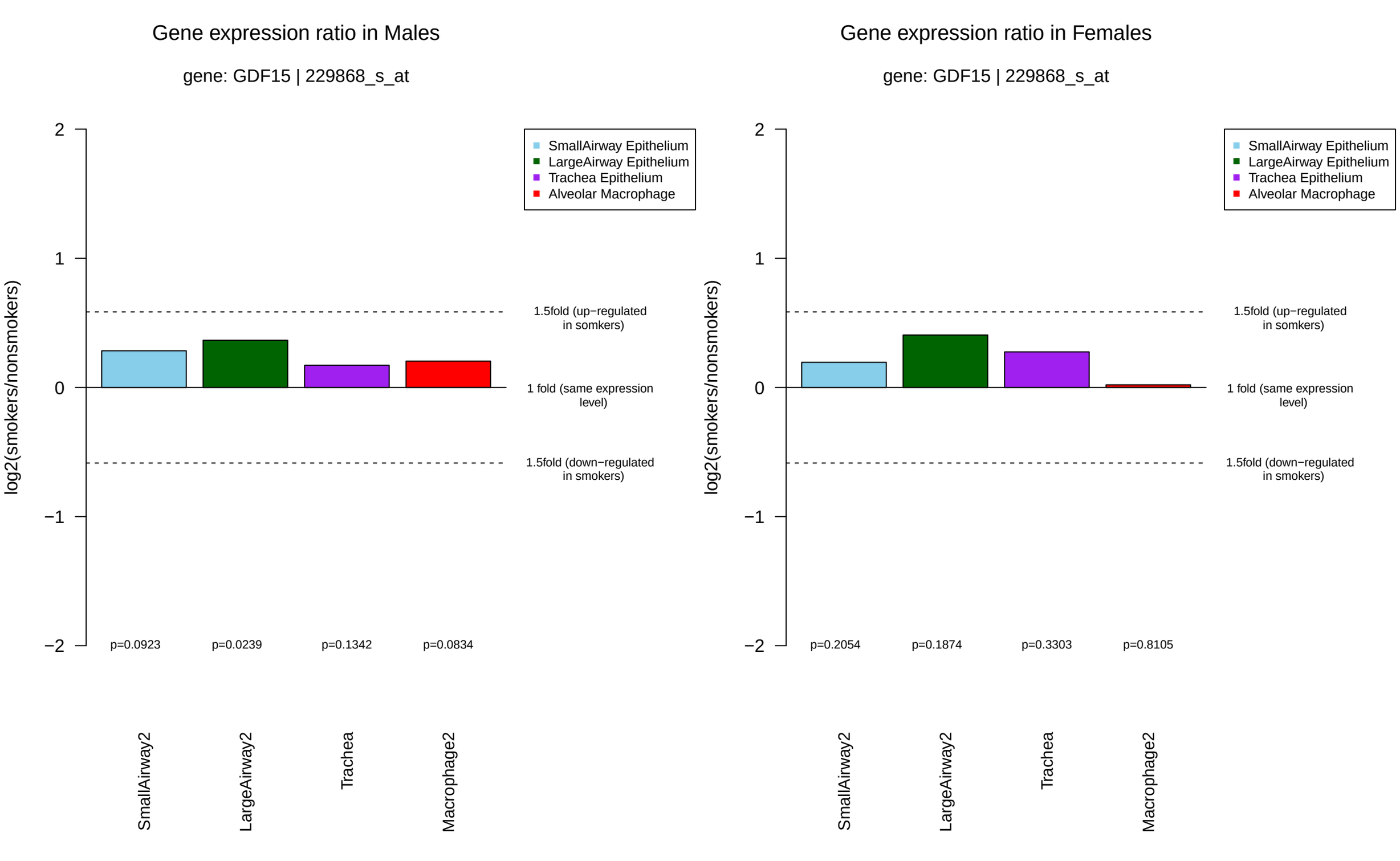
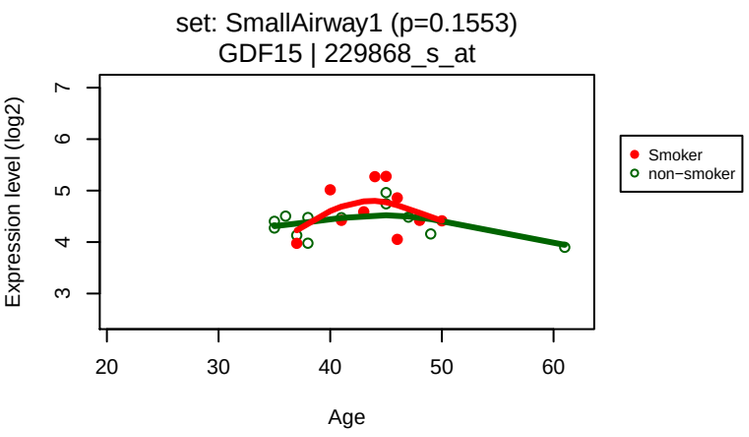
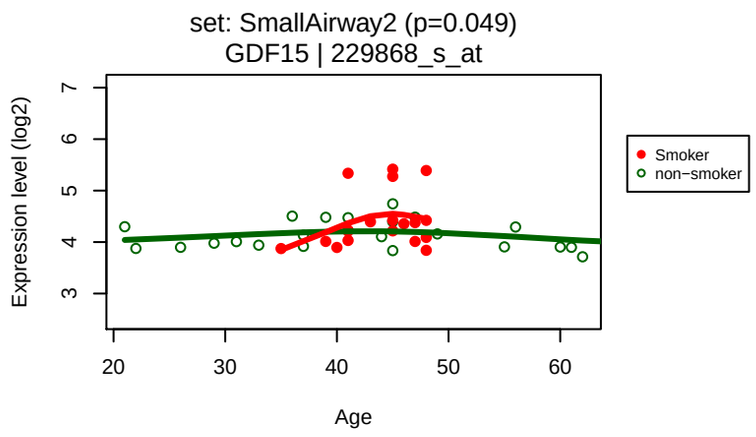
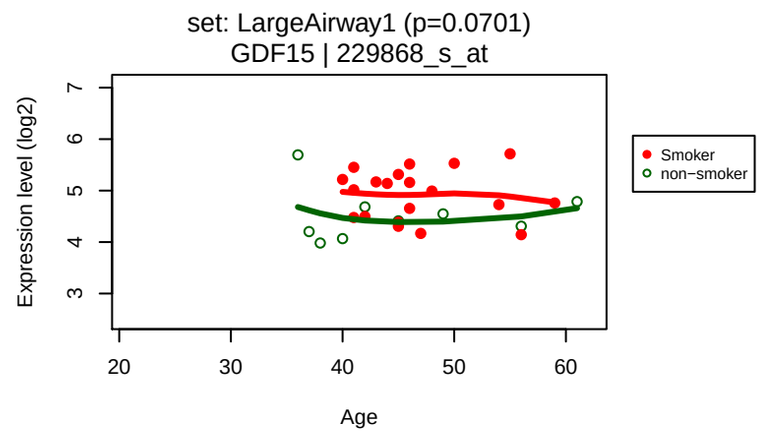

Gene expression of smokers and nonsmokers
gene: GDF15 | 229868_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1553
p=0.049
p=0.0202
p=0.0701
p=0.0223
p=0.0406
p=0.5038
p=0.0867
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

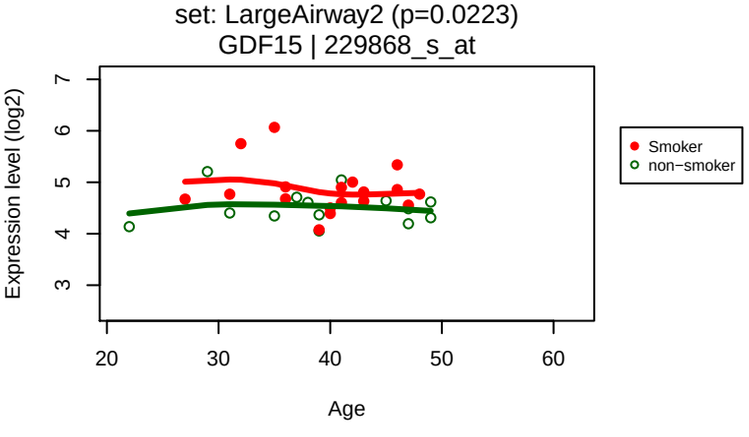

Gene expression ratio (smokers/nonsmokers)
gene: GDF15 | 229868_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1553
p=0.049
p=0.0202
p=0.0701
p=0.0223
p=0.0406
p=0.5038
p=0.0867
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

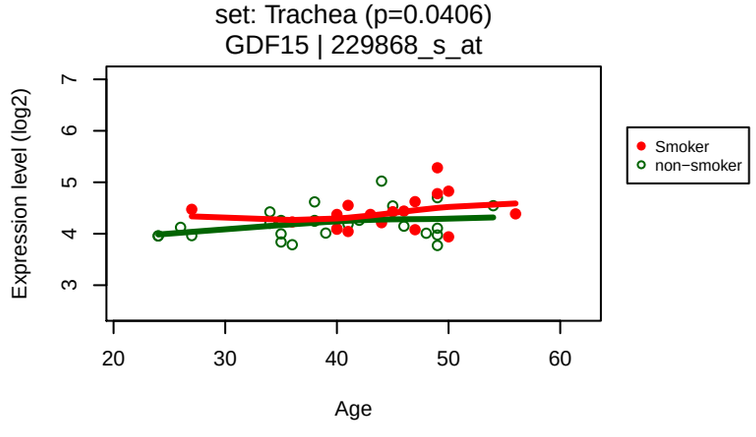

Gene expression ratio in Males
gene: GDF15 | 229868_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0923
p=0.0239
p=0.1342
p=0.0834
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GDF15 | 229868_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2054
p=0.1874
p=0.3303
p=0.8105
SmallAirway2
LargeAirway2
Trachea
Macrophage2

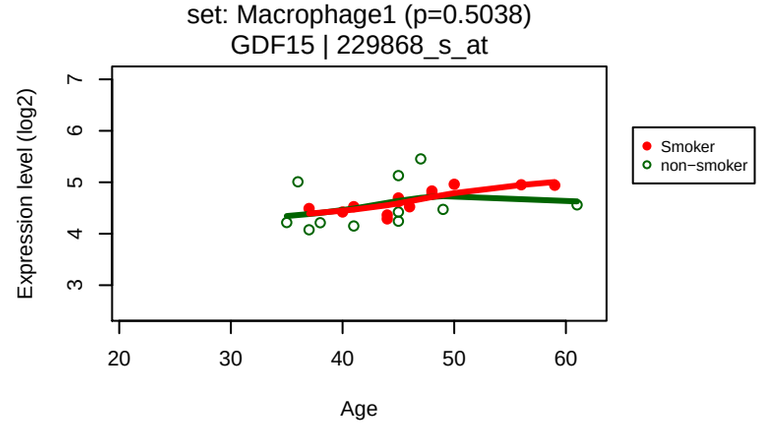

set: SmallAirway1 (p=0.1553)
GDF15 | 229868_s_at
Expression level (log2)
Age
Smoker
non−smoker

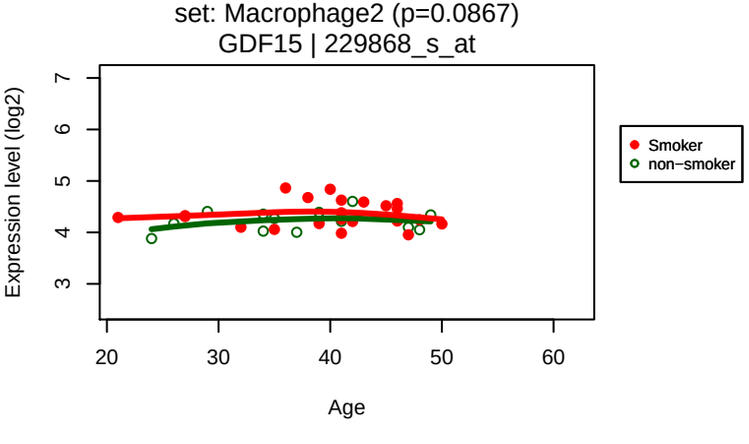

set: SmallAirway2 (p=0.049)
GDF15 | 229868_s_at
Expression level (log2)
Age
Smoker
non−smoker

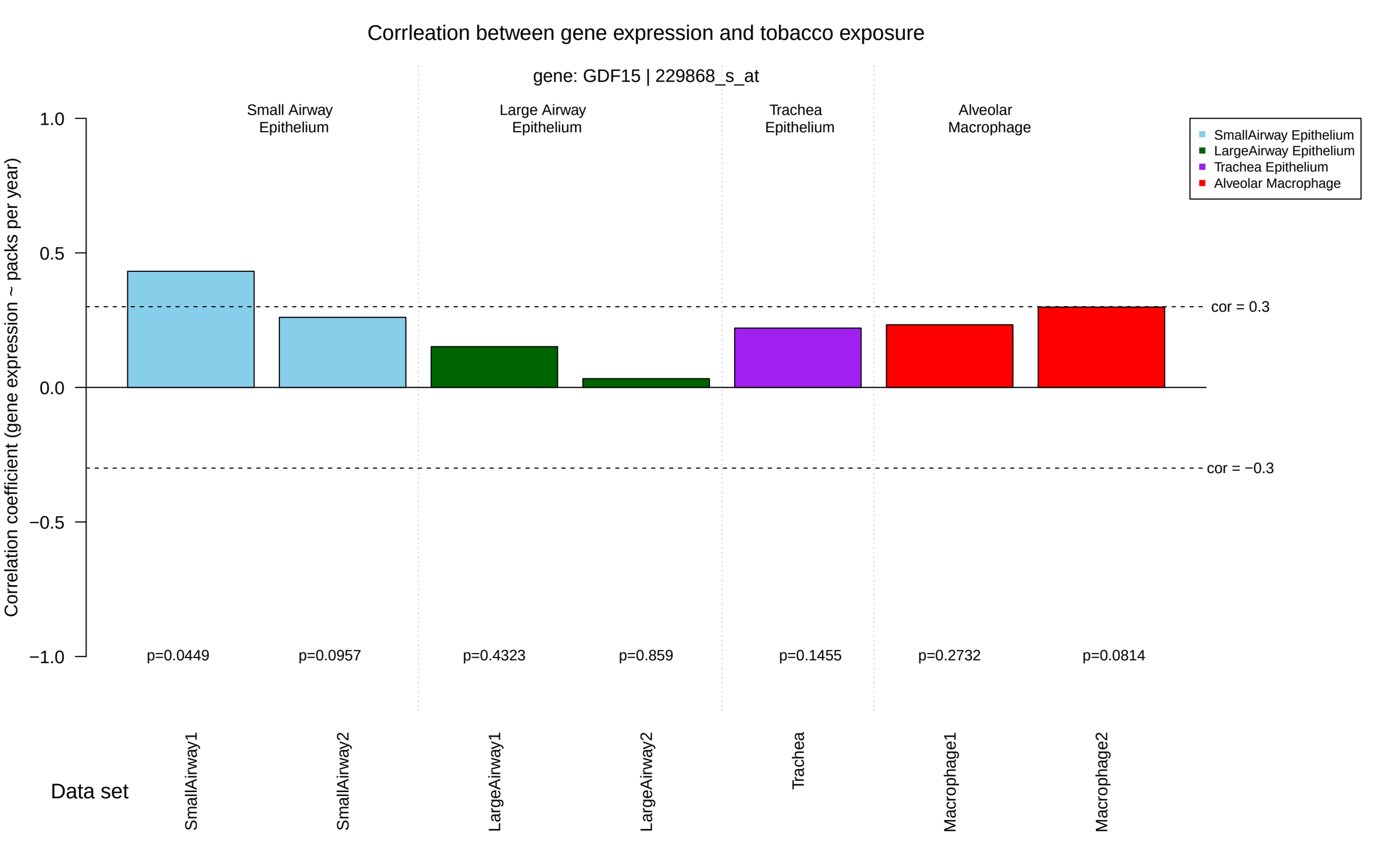

set: LargeAirway1 (p=0.0701)
GDF15 | 229868_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0223)
GDF15 | 229868_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0406)
GDF15 | 229868_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5038)
GDF15 | 229868_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0867)
GDF15 | 229868_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GDF15 | 229868_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0449
p=0.0957
p=0.4323
p=0.859
p=0.1455
p=0.2732
p=0.0814
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2